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#4

SCORING FUNCTIONS

VALIDATING TWIN DRILL HOLES FOR RESOURCE ESTIMATION

Stop guessing and start quantifying.

>>>



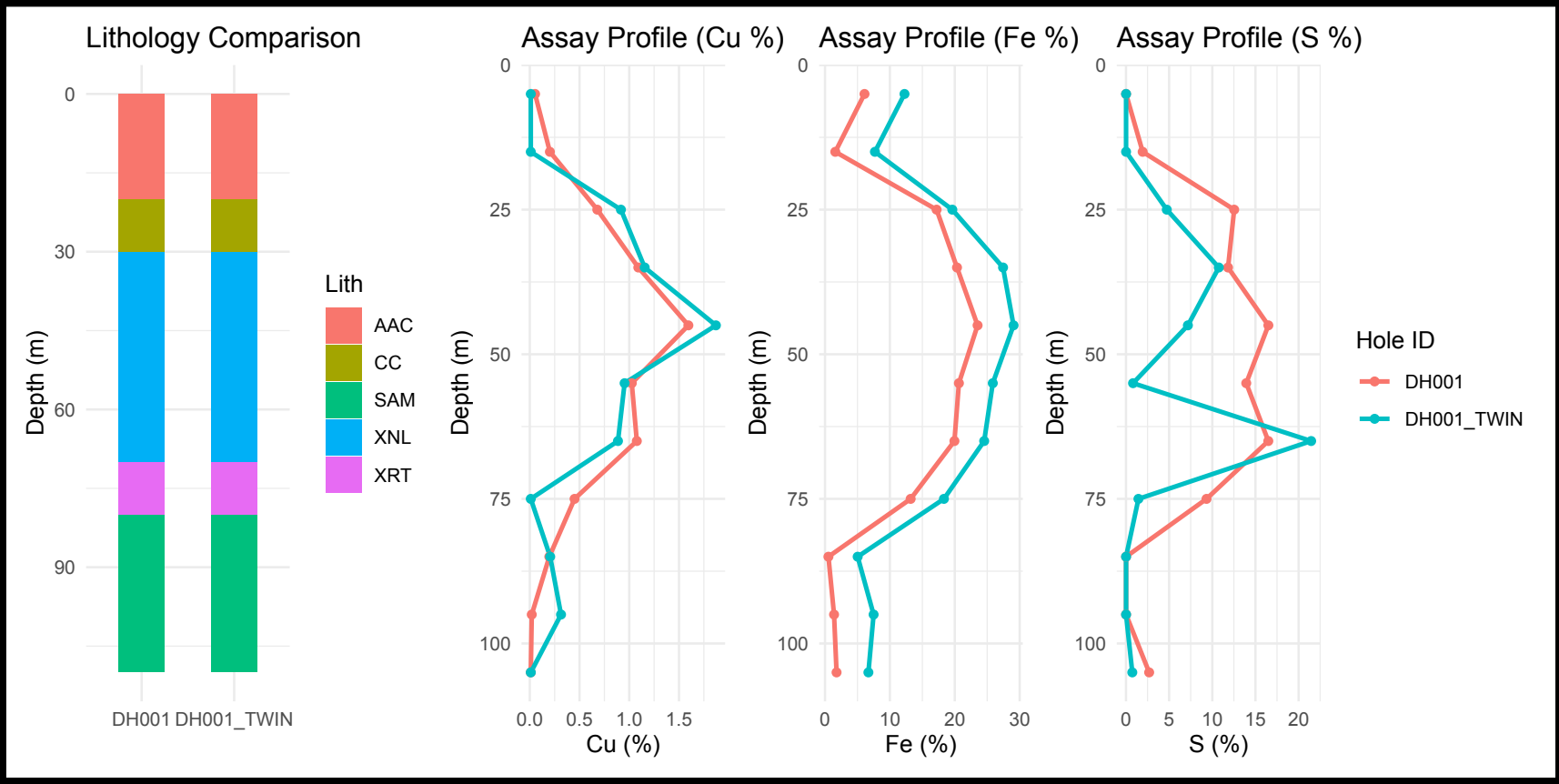
The traditional “Twin Hole” blind spot

Twin drilling is our gold standard for validating historical data. But the traditional review workflow is broken.

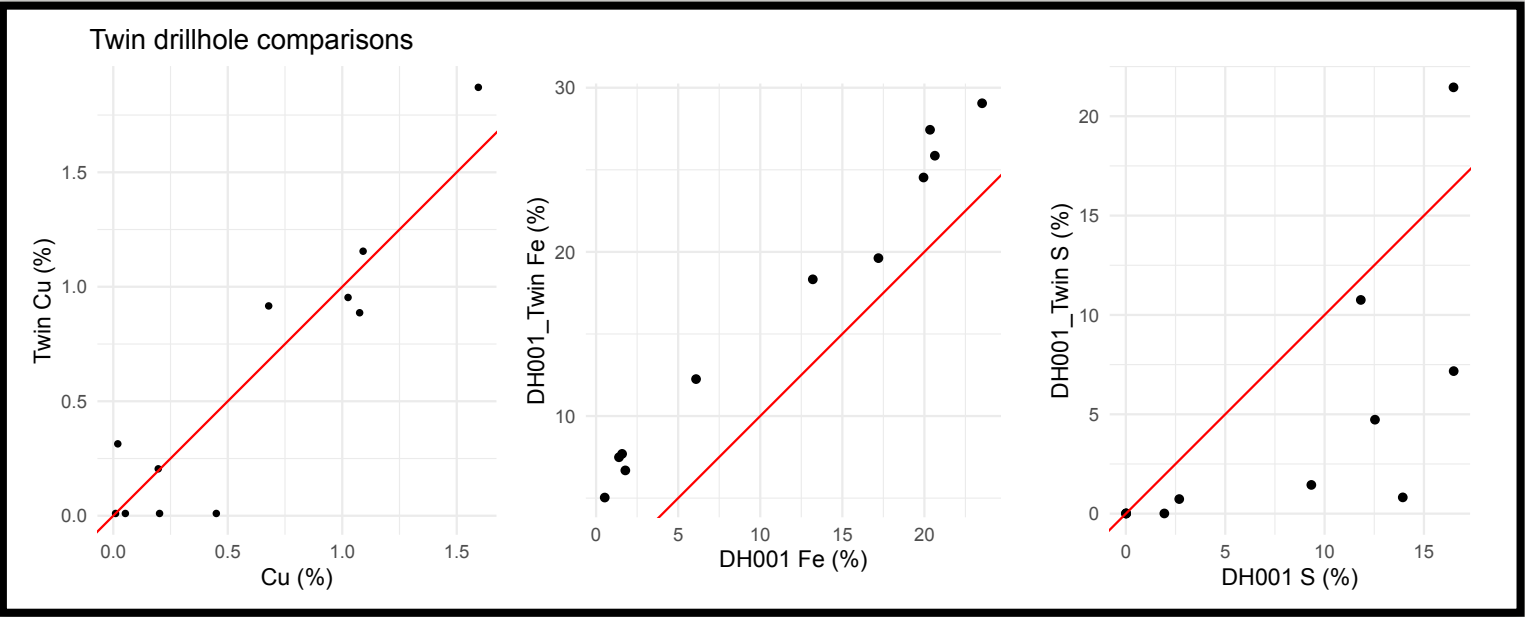
- **It is subjective:**
Plot drill hole traces, squint at scatter plots, and make a judgement.
- **It does not scale well:**
You can visually audit several plots, but what about 50?

>>>

The plots



Drill hole
trace



Scatter plots

>>>

Enter the Scoring Function

A Scoring Function strips away subjectivity by mapping raw physical variance into a standardised 0% to 100% Data Quality Score (DQS).

1. **Flag the problem.**

Run a Max-Normalized Bounded Check to isolate relative difference without low-grade distortions for each assay.

2. **Apply a threshold.**

Turn the relative difference into an indicator.

3. **Calculate prevalence.**

Determine how widespread the problem is.

4. **Penalise logically.**

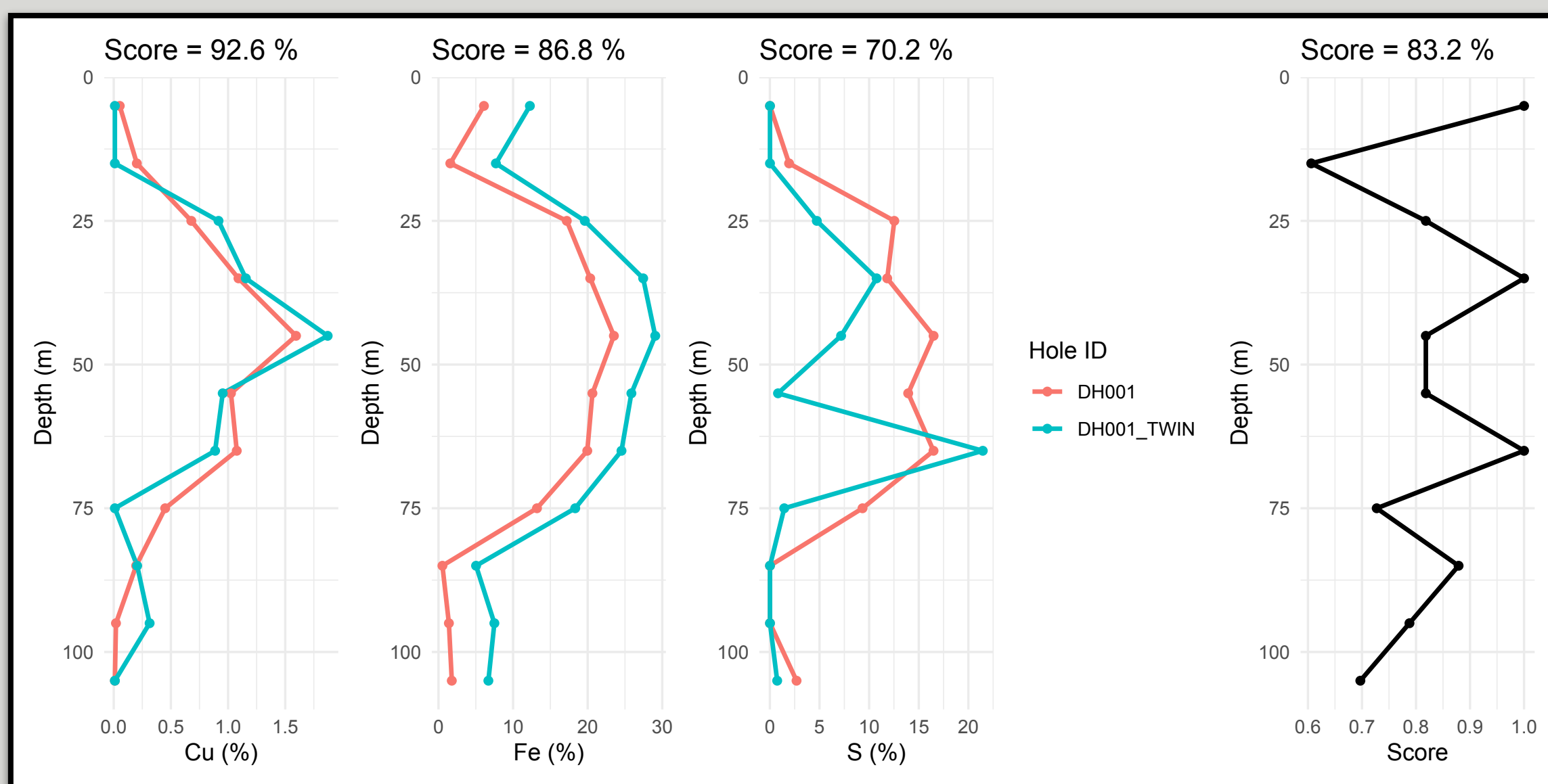
If an interval passes, its score is 1. If it fails, it receives a localised penalty (1-prevalence).

5. **Summarise.**

Calculate the average score by row, by column, by twin drill hole pair, or by dataset.

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The updated plot



With calculated scores.

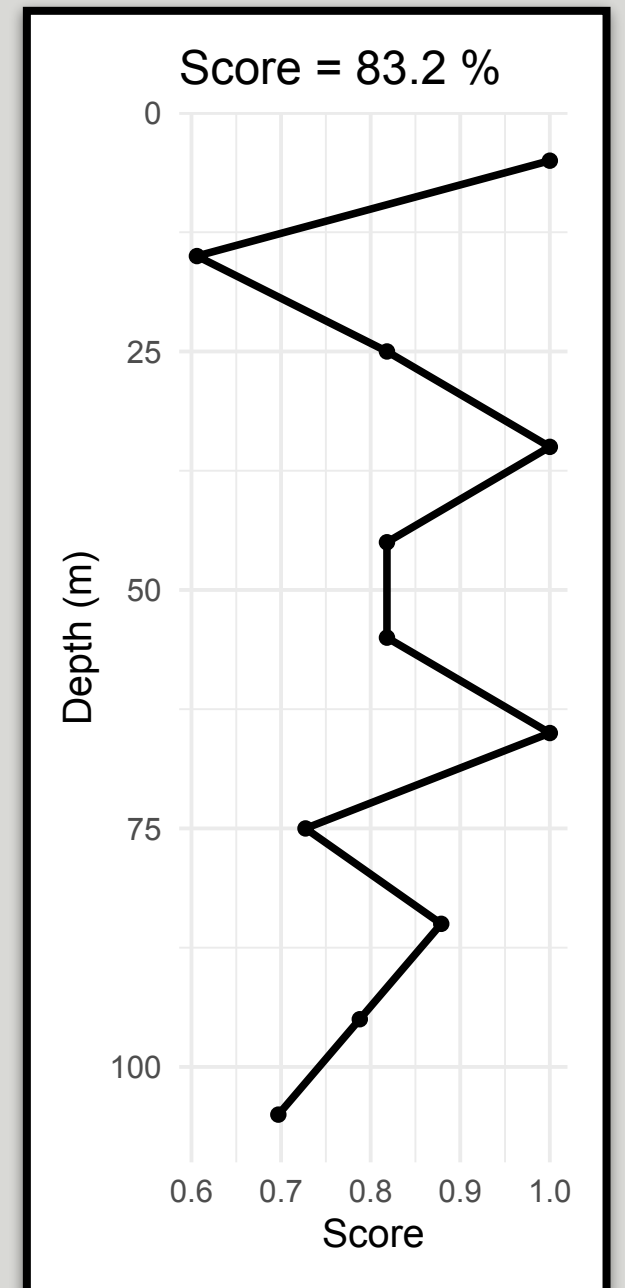
>>>

The SCORE

- The individual scores can be visualised, summarised, and compared.
- They can be used in ML.
- They can be used in spatial estimation.
- They are repeatable, comparable, and deterministic.
- They become a benchmark.

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Scaling the workflow

The scoring function illustrated here is one component of a rigorous, end-to-end data health-check framework that I have been developing.

The framework is designed to assess Geological and Geometallurgical data before resource estimation and predictive modelling.

I deploy this systematic workflow through data profiling and gap analysis.

The goal is simple: eliminate hidden data risk before it corrupts your downstream spatial models.

> > >

The full blueprint is live

This carousel is just the executive summary. The full version can be found on my repository's GitHub Pages.



Interactive HTML Notebook:

Perfect for desktop reading and exploring the code.



Print-Ready PDF Guide: A standalone reference white paper to save locally or share.



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See the code, build the workflow



End-to-end
R/Quarto
blueprint in the
mining-ds-vault

